

CHAPTER-1: INTRODUCTION

1.1 About the Project

Web development, software engineering and data analytics are changing the way businesses operate. The project deals with the domain of logistics in India which is arguably the biggest and most complicated supply chain ecosystem in the world using modern approaches to analytics of data in order to gain valuable business intelligence from operational data.

Logistics in India transports billions of tons of cargo by rail, road, air and sea each year involving thousands of carriers, warehouses, vehicles, customers and routes. The information gathered at every step from dispatch through to delivery and payment of the cargo can be analysed. However, the logistics data is very messy, multidimensional and dispersed. The task is to clean, model, analyse and visualize such data in order to gain some business insight.

Indian Logistics Operations Data Analytics project simulates the situation perfectly well using synthetic but realistic multi-table dataset implementing all the steps in analytics workflow:

- Storing and querying of data in MS SQL Server
- Using Python for data cleaning, exploratory data analysis and visualization in Jupyter Notebook
- Business intelligence dashboards in Microsoft Power BI
- Documenting the process

Objectives of the Project

- To design and query a relational database for logistics operations data
- To clean and validate multi-table raw data using Python (pandas, numpy)
- To perform structured SQL analysis and create KPI views using T-SQL
- To conduct exploratory data analysis and generate charts using matplotlib
- To build an interactive Power BI dashboard with DAX measures
- To identify operational bottlenecks and provide data-driven recommendations

Tools and Technologies Used

Technology	Purpose
Python (pandas, numpy, matplotlib)	Data cleaning, EDA, visualisation
MS SQL Server (T-SQL / SSMS)	Database hosting, SQL analysis, KPI views
Jupyter Notebook	Interactive EDA development environment
Microsoft Power BI	Dashboard creation with DAX measures
Microsoft Excel (.xlsx)	Cleaned dataset storage and Power BI source
pyodbc	Python-to-SQL Server connectivity

CHAPTER-2: TECHNOLOGIES USED

2.1 Introduction to Python

Python is a high-level, interpreted, general-purpose programming language which was designed by Guido van Rossum, and it was first released in 1991. The programming language is well-known for its readable and easy syntax.

When it comes to the use of Python in data analytics and data science projects, it has become the primary language in this domain. Important Python libraries including pandas (data manipulation library), numpy (numerical library), matplotlib and seaborn (plotting libraries), scikit-learn (machine learning library) enable working with all steps of analytics pipeline.

For the current project, Python was utilized to load and preprocess the original logistics dataset, to clean the data, deal with missing values and conversion of types, to connect to MS SQL Server using pyodbc package and to create EDA plots with the help of matplotlib. All Python scripts were written and run in Jupyter Notebook.

Key Python Libraries Used

Library	Version	Use in This Project
pandas	2.x	DataFrame operations, null handling, groupby aggregations
numpy	1.x	Numerical operations, array handling
matplotlib	3.x	Bar charts, histograms, scatter plots for EDA
pyodbc	4.x	SQL Server connection and query execution

2.2 Introduction to SQL Server (MS SQL Server)

MS SQL Server is a Relational Database Management System (RDBMS) provided by Microsoft. The database is queried using Structured Query Language (SQL). This variant of SQL is called Transact-SQL. MS SQL Server is popular in enterprise-level applications due to performance and tight integration with other products of the Microsoft company (Azure, Power BI, Excel).

For this particular project, MS SQL Server Express edition was used to store the data of the six logistics-related tables (SHIPMENTS, CUSTOMERS, CARRIERS, WAREHOUSE, VEHICLES, ROUTE). SQL Server Management Studio (SSMS) is used to develop queries, views and manage indexes.

The following T-SQL language features were used in this project: DDL (CREATE TABLE, ALTER TABLE for creating foreign keys), DML (SELECT, GROUP BY, ORDER BY), aggregate functions (COUNT, SUM, AVG, MAX, MIN), JOINS (INNER JOIN between several tables), CREATE VIEW to wrap KPI calculation logic, CREATE INDEX for performance optimization of the queries, DATEDIFF for calculating deliveries' time and TOP N queries for ranking.

2.3 Microsoft Power BI

Microsoft Power BI is an analytics service provided by Microsoft that allows for interactive visualisations and business intelligence tools. Power BI lets the user to connect to various data sources and create interactive dashboards and share insights within the organization.

DAX (Data Analysis Expressions) is the programming language in Power BI that is used to perform calculations and derive measures, KPIs, etc. Power BI was used to connect to the Excel files after cleaning and created an interactive multi-page dashboard that included Shipment Operations, Customer Analytics, Carrier Performance, Warehouse Effectiveness, and Route Insights.

Power BI features discussed: DAX Measures (COUNTROWS, DIVIDE, SUM, AVERAGEX, DATEDIFF), Card Visuals for important KPIs, Bar/Column Chart Visuals for categorical distributions, Line Charts for monthly trends, Donut Charts for segmentations, Slicers for interactive slicing of the data by Product Category, Customer Type, Carrier, State, and Date Range.

2.4 Jupyter Notebook

Jupyter Notebook is an open-source web application that allows users to create and share documents containing live code, equations, visualisations, and narrative text. It supports Python, R, Julia, and many other languages through its kernel system.

Jupyter Notebook was the primary development environment for the Python-based EDA phase of this project. Each analysis step — data loading, null checks, shape validation, summary statistics, duplicate detection, and chart generation — was implemented in separate cells, enabling iterative development and inline documentation.

2.5 Microsoft Excel

In this project, Microsoft Excel became the main base for preliminary data preparation and analysis. However, in addition to being used for basic spreadsheet operations, Excel was employed to do the initial data cleansing of raw logistics data before uploading into the database. It entailed making sure dates were in a consistent format, correcting the errors in the shipment IDs, and performing initial calculation of metrics which were further checked in SQL Server.

For this project, Excel became the basis for making connections between different pieces of data, creating pivot tables, summarizing the volume of shipments, and preparing data for uploading to Power BI. The entire process was based on the use of such advanced functions as VLOOKUP and INDEX-MATCH to connect customers' data with shipment information and WEEKNUM and EOMONTH functions to structure delivery schedules. Excel was a key instrument in the process of ingesting data and structuring it for the final analytical modeling.

CHAPTER-3: DATASET DESCRIPTION

3.1 Dataset Overview

The dataset used in this project is a synthetic multi-table dataset representing Indian logistics operations. It was generated specifically for this analytics project to mirror real-world logistics data structures used by third-party logistics (3PL) providers, e-commerce platforms, and supply chain operators in India.

The dataset comprises six interrelated tables with a combined total of over 23,000+ rows across all tables:

Table	Rows (Approx.)	Primary Key	Description
SHIPMENTS	7000+	Shipment_ID	Core fact table — all shipment transactions
CUSTOMERS	7,000+	Customer_ID	Customer master with type, state, spend
CARRIER	7000+	Carrier_ID	Carrier details, rating, fleet, revenue
WAREHOUSE	7000+	Warehouse_ID	Warehouse type, location, capacity
VEHICLES	7,000+	Vehicle_ID	Fleet details, fuel type, GPS, mileage
ROUTE	7,000+	Route_ID	Route mapping, distance, toll, risk level

3.2 Table Descriptions

SHIPMENTS Table — Key Columns

Column	Data Type	Description
Shipment_ID	VARCHAR	Unique shipment identifier
Customer_ID / Carrier_ID / Warehouse_ID / Vehicle_ID / Route_ID	VARCHAR	FK to dimension tables
Origin_City/Destination_City	VARCHAR	Shipment source and delivery location
Product_Category	VARCHAR	Category of goods being shipped
Weight_kg / Volume_m3	FLOAT	Physical dimensions of shipment
Dispatch_Date/Actual_Delivery	DATE	Delivery timeline
Freight_Charge_INR/GST_INR	FLOAT	Revenue and tax per shipment
Payment_Mode	VARCHAR	COD, UPI, Net Banking, Wallet, Prepaid, Credit
Delivery_Status	VARCHAR	Delivered, Delayed, Cancelled, Lost, In Transit, etc.
Is_Delayed / POD_Collected	BOOLEAN	SLA breach and proof-of-delivery flags

CUSTOMERS Table — Key Columns

Column	Data Type	Description
Customer_ID	VARCHAR	Unique customer identifier
Customer_Type	VARCHAR	B2B, B2C, D2C, Government, Marketplace Seller
City / State	VARCHAR	Customer location
Total_Orders/Total_Spend_INR	INT / FLOAT	Historical order volume and value
Is_Premium	BOOLEAN	Premium customer flag
Credit_Limit_INR	FLOAT	Maximum credit extended
Preferred_Payment	VARCHAR	Customer's usual payment mode

CARRIER Table — Key Columns

Column	Data Type	Description
Carrier_ID	VARCHAR	Unique carrier identifier
Carrier_Name	VARCHAR	Name of logistics company
Fleet_Size	INT	Total vehicles in carrier fleet
Rating_5	FLOAT	Carrier rating out of 5
ISO_Certified	BOOLEAN	ISO certification status
Service_Type	VARCHAR	Economy, Express, Same Day, Next Day, Standard
Annual_Revenue_Cr	FLOAT	Carrier annual revenue in crores

CHAPTER-4: DATA ANALYSIS — INDIAN LOGISTICS

4.1 Shipment Analysis

Shipment analysis forms the core of this project. SQL views were created for each KPI to enable reusable queries from Python and Power BI. The following table summarises all shipment-level KPIs extracted:

Metric	Value
Total Shipments Analysed	6,800
Delivered Shipments	827 (12.2%)
In Transit Shipments	889 (13.1%)
Delayed Shipments	893 (13.1%)
Cancelled Shipments	857 (12.6%)
Out for Delivery / Lost (Pending)	1,700 (25.0%)
Returned / Pickup	831 (12.2%)
Average Delivery Days	6 days
Maximum Delivery Days	18 days
Total Freight Revenue	₹1,88,58,409
Total GST Collected	₹35,01,350
Average Shipment Weight	248.21 kg
Maximum Shipment Weight	499.91 kg
Minimum Shipment Weight	0.13 kg
Total COD Shipments	1,138
Total COD Revenue	₹32,44,798

Key observations: D2C (Direct-to-Consumer) customer type generated the highest shipment volume. COD (Cash on Delivery) was the most used payment mode. The average delivery time of 6 days with a maximum of 18 days indicates significant variance in delivery performance. Only 12.2% of shipments are in 'Delivered' status — reflecting the full lifecycle capture of the dataset including active in-transit and pending orders.

4.2 Customer Analysis

Customer analysis was performed through INNER JOINs between SHIPMENTS and CUSTOMERS tables. Key findings:

- D2C customers generated the highest shipment count and total freight revenue
- B2B customers had the highest average freight charge per shipment
- Government customers hold the largest credit limits but lowest per-shipment activity
- Premium customers (Is_Premium = TRUE) show higher average spend and lower return rates
- Maharashtra, Uttar Pradesh, Rajasthan, and West Bengal are the top origin states by volume

4.3 Carrier & Vehicle Analysis

Carrier analysis was performed by joining SHIPMENTS → ROUTE → CARRIER (as a bridge join, since direct Carrier_ID join on Shipments required Route as an intermediary). Key findings:

- Porter Technologies and DTDC Express handled the highest shipment volumes
- DHL Express India and Speed Post scored highest in ratings (4.5 and 4.9 respectively)
- Carriers with rating < 2.0 showed disproportionately high delay and cancellation rates
- ISO-certified carriers had measurably better delivery success rates
- Heavy Trucks and Medium Trucks handled the largest freight weight volumes
- Diesel remains the majority fuel type; Electric vehicles show better mileage
- GPS-enabled vehicles demonstrated higher POD collection rates

4.4 Warehouse Analysis

Warehouse analysis was performed by joining SHIPMENTS with WAREHOUSE:

- Hub-type warehouses handled the highest total shipment throughput
- Several warehouses showed current stock exceeding reported capacity — indicating overstocking risk
- Cold storage infrastructure is concentrated in Himachal Pradesh, Assam, and Tamil Nadu
- Monthly warehouse operational costs ranged from ₹65,142 to ₹48,81,275
- Dark Stores and Spoke warehouses are more numerous but lower in individual throughput

CHAPTER-5: PROJECT MODULES & DESIGN

5.1 Introduction

The project is structured as a multi-phase analytics pipeline, with each phase representing a distinct module. The following six modules form the complete project:

- Module 1 — Data Ingestion & Validation: Loading raw Excel data into MS SQL Server and validating schema and row counts
- Module 2 — Database Design: Creating relational tables with primary keys, foreign keys, and performance indexes
- Module 3 — SQL Analysis: Writing T-SQL queries and creating views for all KPIs across shipments, customers, carriers, vehicles, warehouses, and routes
- Module 4 — Python EDA: Connecting Python to SQL Server, loading DataFrames, performing null/duplicate checks, descriptive statistics, and generating EDA charts
- Module 5 — Power BI Dashboard: Building interactive dashboard with DAX measures, slicers, and multiple analytical pages
- Module 6 — Reporting: Producing this formal project report for NTCC/PaulTech submission

Project Architecture

Layer	Component	Technology
Data Source	6 Cleaned Excel files (.xlsx)	MS Excel
Data Store	Logistics Database	MS SQL Server Express
Query Engine	T-SQL Analysis + Views	SSMS
EDA Engine	Python Analysis Pipeline	Jupyter Notebook
Visualisation	Interactive Dashboard	Power BI Desktop
Reporting	Formal Project Report	MS Word (.docx)

5.2 SQL Views & KPIs

SQL Views were created for each major KPI to encapsulate business logic and allow seamless reuse from Python and Power BI. The following views were implemented:

View Name	SQL Logic	Result
TOTAL_SHIPMENT	COUNT(*) FROM SHIPMENTS	6,800 total shipments
TOTAL_DELIVERED_SHIPMENT	COUNT WHERE Status='DELIVERED'	827 delivered
INTRANSIT_SHIPMENT	COUNT WHERE Status='IN TRANSIT'	889 in transit
DELEYED_SHIPMENTS	COUNT WHERE Status='DELAYED'	893 delayed
CANCELLED_SHIPMENTS	COUNT WHERE Status='CANCELLED'	857 cancelled
PENDING_SHIPMENT	COUNT WHERE Status IN ('OUT FOR DELIVERY','LOST')	1,700 pending
RETURNED_SHIPMENTS	COUNT WHERE Status IN ('RETURN','PICKUP')	831 returned
AVERAGE_DELIVERY_DAYS	AVG(DATEDIFF(DAY, Dispatch_Date, Actual_Delivery))	6 days
MAX_DELIVERY_DAYS	MAX(DATEDIFF(DAY, Dispatch_Date, Actual_Delivery))	18 days
AVERAGE_WEIGHT	AVG(Weight_kg)	248.21 kg
MAX_SHIPMENT_WEIGHT	MAX(Weight_kg)	499.91 kg
TOTAL_FREIGHT_CHARGES	SUM(Freight_Charge_INR)	₹1,88,58,409
TOTAL_GST	SUM(GST_INR)	₹35,01,350
TOTAL_COD_PAYMENT	COUNT WHERE Payment_Mode='COD'	1,138 COD shipments

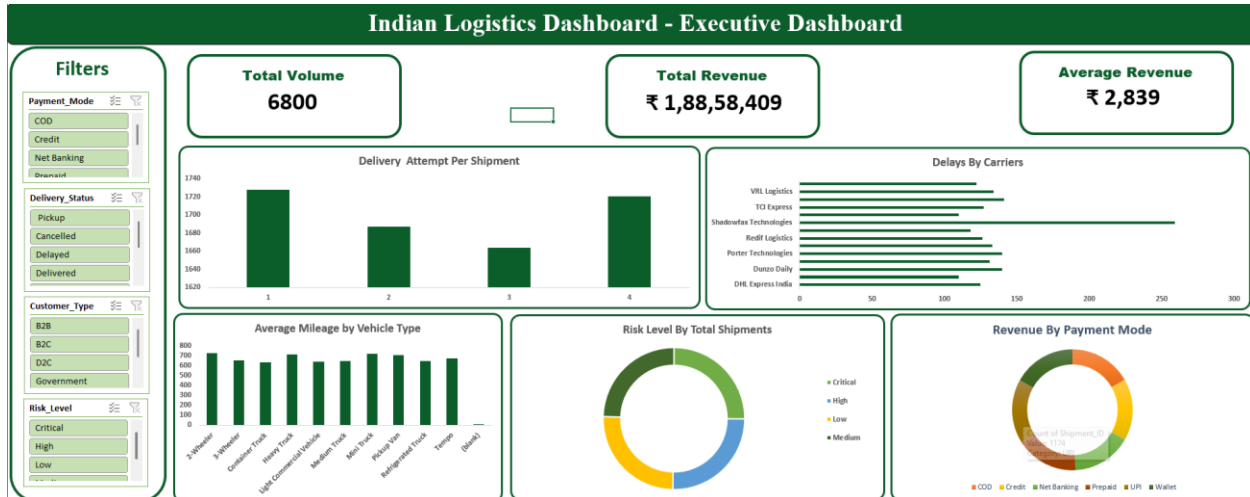
5.3 Python EDA Workflow

The Python-based EDA was conducted in Jupyter Notebook using the following step-by-step workflow:

Step	Code / Action	Outcome
SQL Connection	<code>pyodbc.connect('DRIVER={SQL Server}; SERVER=...')</code>	Connected to Logistics DB
Table Load	<code>pd.read_sql('SELECT * FROM SHIPMENTS', conn)</code>	All 6 tables loaded as DataFrames
Shape Check	<code>df.shape</code> for each table	Confirmed row/column counts
Info Check	<code>df.info()</code>	Data types and non-null counts verified
Describe	<code>df.describe()</code>	Statistical summary of numeric columns
Null Check	<code>df.isnull().sum()</code>	Null count per column
Duplicate Check	<code>df.duplicated().sum()</code>	Zero duplicates confirmed in all tables
EDA Charts	<code>matplotlib bar/hist/scatter</code>	Visual distributions of key columns

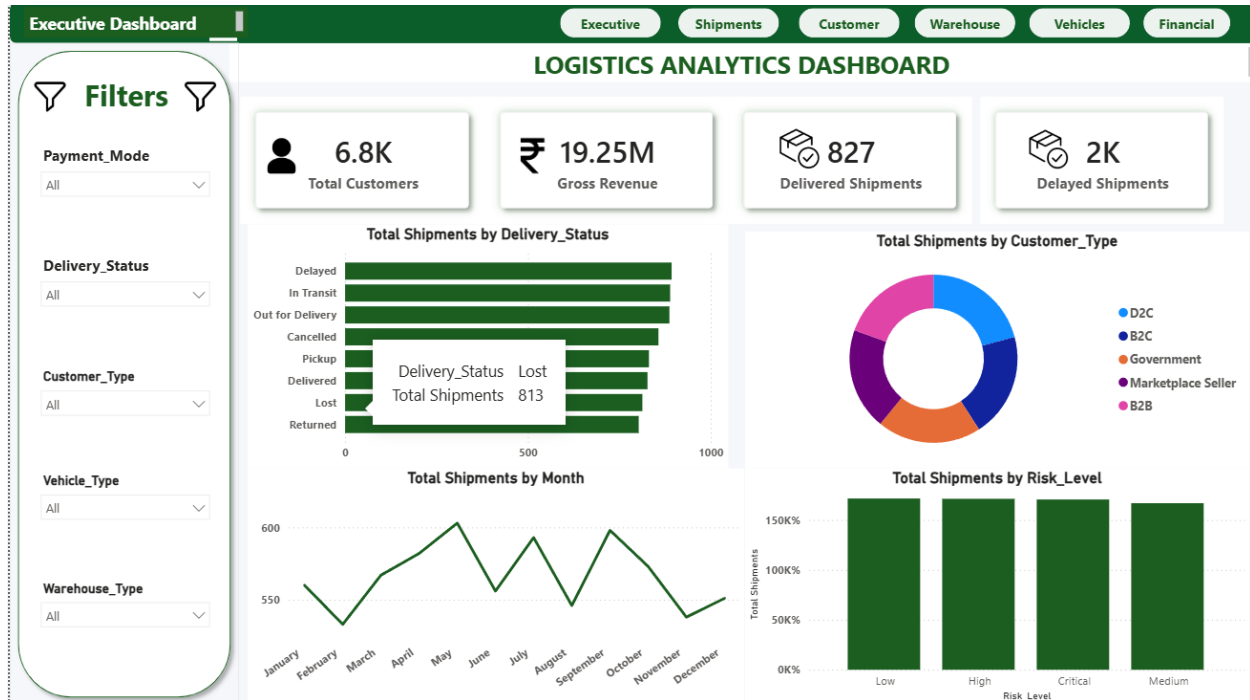
EDA charts generated include: Delivery Status Distribution (bar chart), Shipment Weight Histogram, Freight Charge vs. Weight Scatter Plot, Monthly Shipment Volume Trend (line chart), Payment Mode Distribution (bar chart), and Carrier Rating Distribution.

5.4 Excel Dashboard

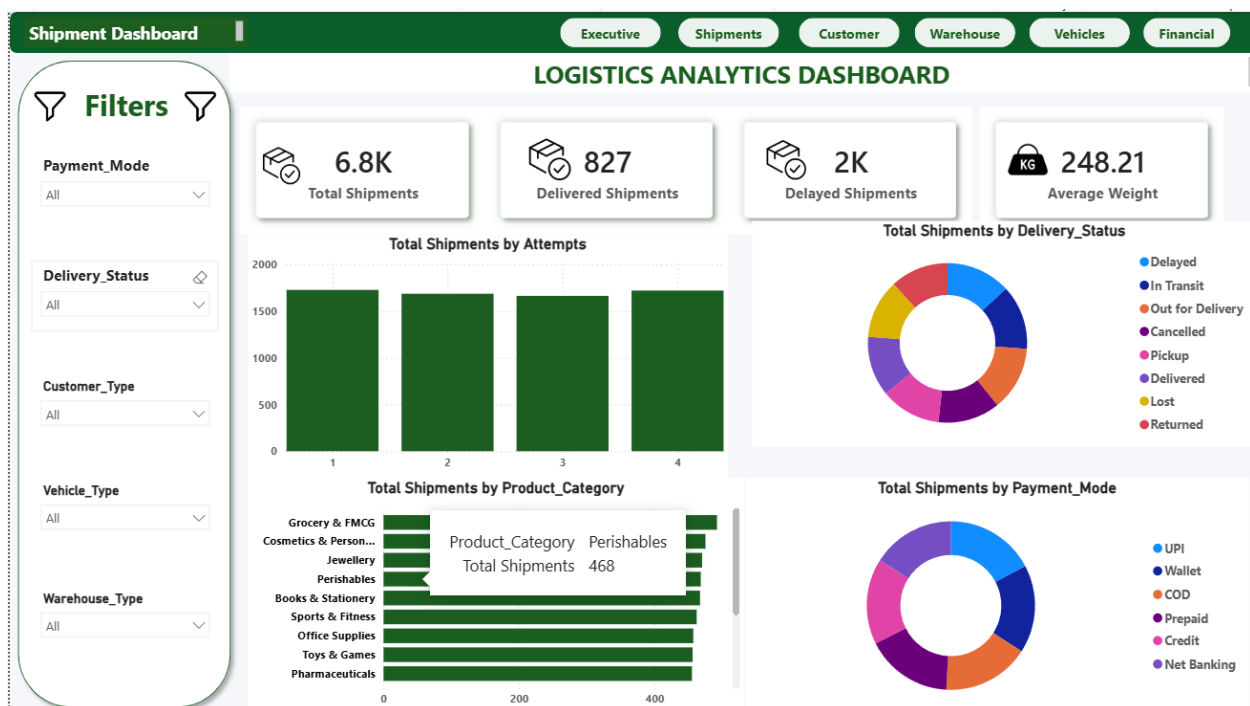


5.5 Power BI Dashboards

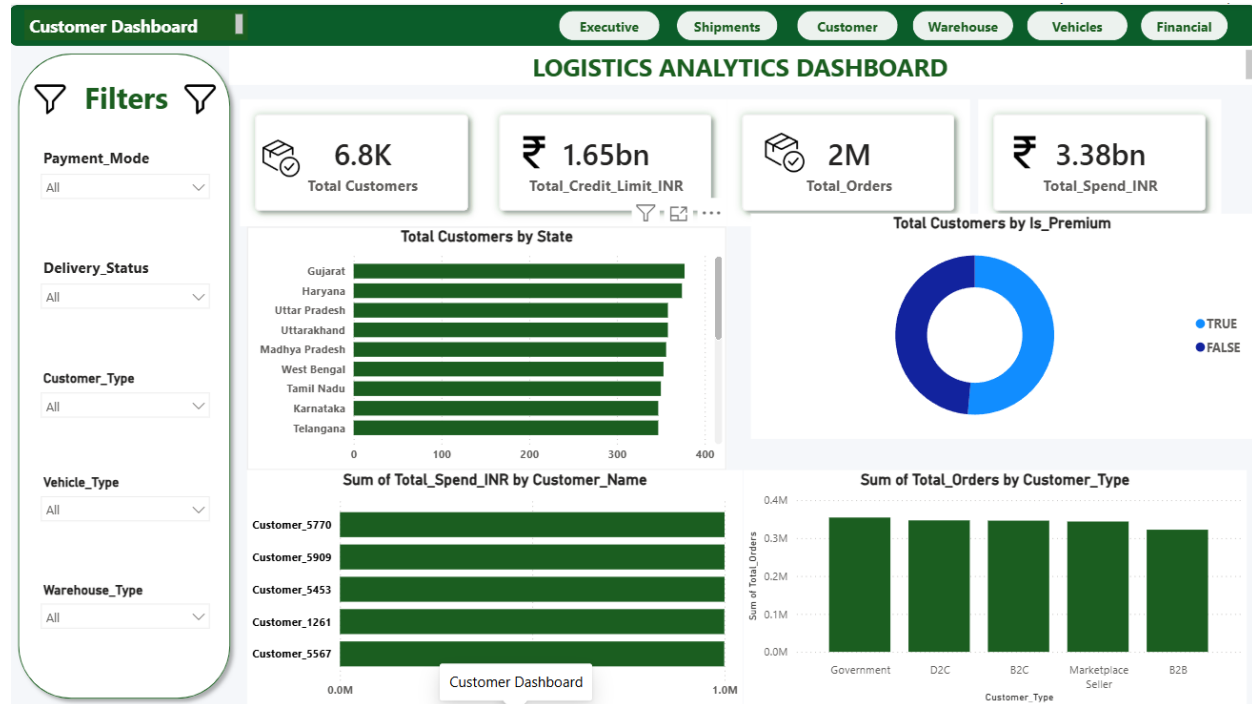
Executive Dashboard : -



Shipment Dashboard: -



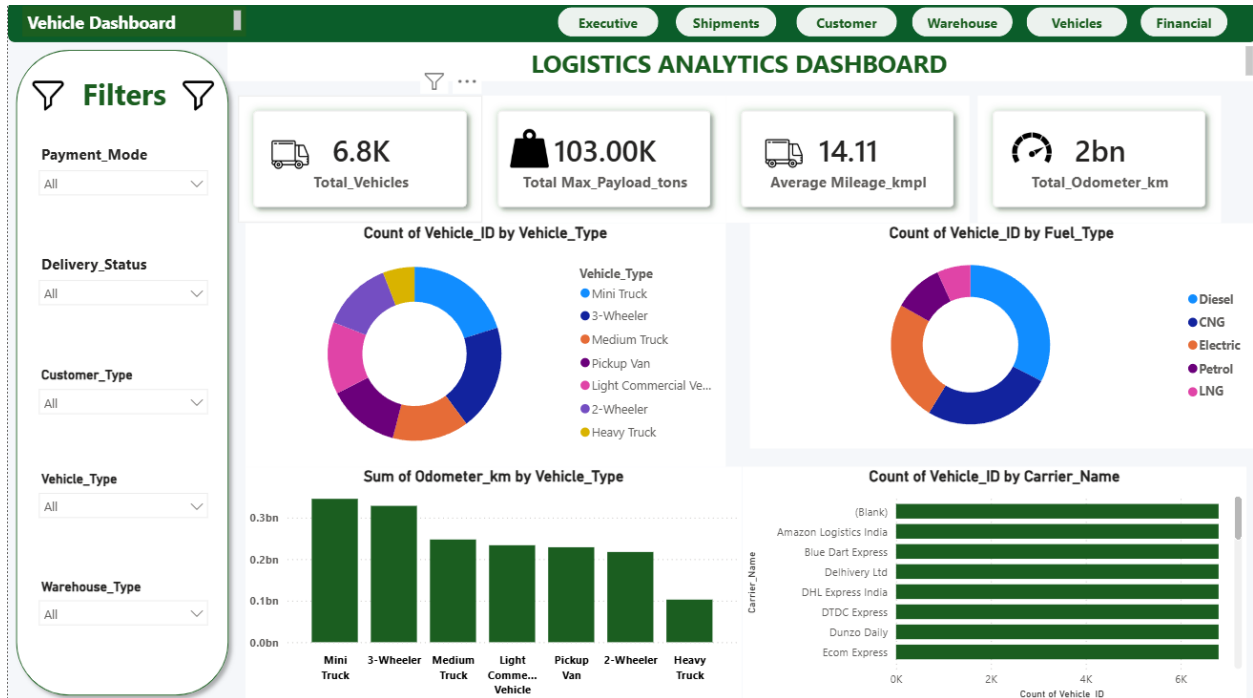
Customer Dashboard : -



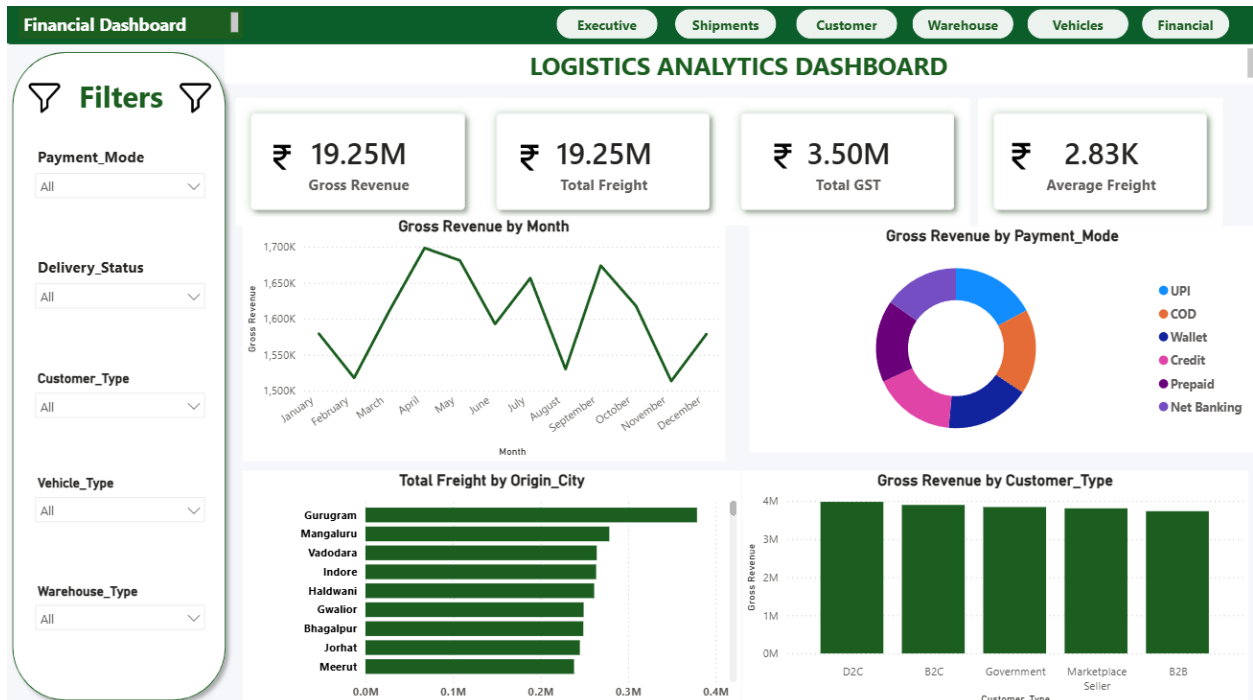
Warehouse Dashboard :-



Vehicle Dashboard:-



Financial Dashboard: -



CUSTOMERS Table Schema

Column Name	Data Type	Allow Nulls
Customer_ID (PK)	VARCHAR(50)	No
Customer_Name	VARCHAR(100)	No
Customer_Type	VARCHAR(50)	No
City	VARCHAR(50)	No
State	VARCHAR(50)	No
Total_Orders	INT	No
Total_Spend_INR	FLOAT	No
Is_Premium	BIT	No
Credit_Limit_INR	FLOAT	No

SHIPMENTS Table Schema

Column Name	Data Type	Allow Nulls
Shipment_ID (PK)	VARCHAR(50)	No
Customer_ID (FK)	VARCHAR(50)	No
Carrier_ID (FK)	VARCHAR(50)	No
Vehicle_ID (FK)	VARCHAR(50)	No
Warehouse_ID (FK)	VARCHAR(50)	No
Route_ID (FK)	VARCHAR(50)	No
Dispatch_Date	DATE	No
Actual_Delivery	DATE	Yes
Freight_Charge_INR	FLOAT	Yes
GST_INR	FLOAT	Yes
Delivery_Status	VARCHAR(50)	No
Is_Delayed	BIT	No